The Cost of Alignment: Preserving Factual Reliability in Safety Fine-Tuning

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Abstract:

Recent advancements in Large Language Models (LLMs) have highlighted a critical trade-off between safety alignment and model utility, often termed the "Safety Tax." While recent work by Xue et al. (2025) identifies Multi-Layer Perceptron (MLP) layers as the optimal target for Low-Rank Adaptation (LoRA) to preserve reasoning capabilities, this strategy overlooks mechanistic interpretability research—such as Geva et al. (2021)—suggesting MLPs function as "knowledge warehouses." In this work, we hypothesized that targeting Attention layers for safety alignment would better preserve factual reliability by avoiding interference with these storage mechanisms. To empirically test this, we selected the DeepSeek-R1-Distill-Llama-8B model, intentionally matching one of the models evaluated by Xue et al. This experimental design allowed us to first replicate their findings regarding the preservation of reasoning capabilities, and subsequently evaluate the limitations of their approach regarding factual knowledge.

Contrary to our hypothesis, targeting Attention layers yielded no comparative advantage in preserving factual knowledge; in fact, the MLP baseline demonstrated slightly superior performance. While we anticipated that absolute performance on general knowledge tasks would remain low—given that DeepSeek-R1-Distill-Llama-8B is heavily optimized for mathematical reasoning rather than broad encyclopedic retrieval—we nonetheless expected the Attention configuration to outperform the MLP targeted approach in this domain. The lack of such a margin suggests that the observed low factual accuracy is primarily attributable to the model's pre-existing limitations rather than solely to alignment-induced degradation, regardless of the specific LoRA configuration used.

Introduction

The rapid deployment of Large Language Models (LLMs) in sensitive domains has made safety alignment ensuring models refuse harmful instructions, a critical requirement. However, this alignment process often comes at a cost, a phenomenon widely referred to as the "Safety Tax". Techniques such as Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) can inadvertently degrade a model's utility, particularly its capabilities in complex reasoning and factual recall.

Recent research has sought to minimize this tax by targeting specific components of the Transformer architecture. Notably, Xue et al. proposed that applying LoRA specifically to MLP layers is the optimal strategy for safety alignment. Their work demonstrated that this method effectively induces safety behaviors while preserving mathematical and coding reasoning better than full-parameter tuning.

However, this strategy introduces a theoretical tension. While MLP-targeted adaptation effectively preserves reasoning capabilities, it appears to overlook insights from mechanistic interpretability research. Specifically, Geva et al. (2021) characterize MLP layers as 'Key-Value Memories'—the primary warehouses of a model's factual knowledge. This raises a critical question: Does modifying these 'knowledge warehouses' to instill safety alignment unintentionally compromise the model's factual reliability? We hypothesized that while MLP-focused fine-tuning might preserve procedural reasoning, it could induce 'catastrophic forgetting' of declarative, factual knowledge.

In this project, we investigate this trade-off by fine-tuning the DeepSeek-R1-Distill -Llama-8B model using two distinct configurations. One targeting MLP layers (the baseline) and one targeting Attention layers. We proposed that updating Attention layers, which govern information routing rather than storage, would offer a more robust alternative, preserving both reasoning and factual integrity.

Our experiments reveal that while attention based targeting effectively preserves reasoning capabilities, it fails to offer the hypothesized protection for factual knowledge. Both configurations suffered comparable degradation in general knowledge benchmarks, suggesting that the "Safety Tax" is a systemic fragility in distilled reasoning models rather than a localized defect in specific model layers.

Related work

Our research sits at the intersection of safety alignment, parameter-efficient fine-tuning (PEFT), and mechanistic interpretability. Below, we discuss the existing literature and distinguish our specific contribution.

The "Safety Tax" and Efficient Alignment

The tension between making models safe and preserving their utility is well-documented. Early approaches using Reinforcement Learning from Human Feedback (RLHF) often resulted in the "alignment tax," where models became less capable on downstream tasks (Ouyang et al., 2022).

More recently, the focus has shifted to Low Rank Adaptation as a less destructive alternative to full fine-tuning. The primary reference for our work is Xue et al., "LoRA is All You Need for Safety Alignment of Reasoning LLMs". They conducted extensive experiments showing that:

1. LoRA can achieve safety performance comparable to full fine-tuning.
2. Targeting MLP layers (specifically the up_proj matrix) is the optimal strategy for preserving reasoning capabilities (e.g., coding and math) compared to targeting Attention layers.
3. Rank r=1 is sufficient for these updates.

How we differ: While Xue et al. optimized for reasoning metrics (GSM8K, HumanEval), they did not explicitly evaluate the impact of their method on declarative knowledge. Our work specifically investigates this overlooked dimension.

DeepSeek-R1 and Distilled Reasoning

We exploit DeepSeek-R1-Distill-Llama-8B, a model optimized for reasoning via reinforcement learning and distillation. The DeepSeek Technical Report (2025) highlights that while these models excel at Chain of Thought tasks, they can suffer from hallucinations or gaps in general world knowledge compared to their larger teacher models . **How we differ:** Most safety research treats utility as a single metric. By using DeepSeek-R1, we are able to disentangle utility into two distinct axes: Reasoning (where the model is strong) vs. Knowledge (where it is fragile). Our work tests whether safety fine-tuning exacerbates the model's inherent weakness in general knowledge.

Data

To ensure our findings were directly comparable to the baseline established by Xue et al., we adhered to a strict "apples-to-apples" methodology regarding data selection and preprocessing.

Source and Composition
We utilized the “DirectRefusal” dataset (specifically the TianshengHuang/DirectRefusal repository on HuggingFace)—the same dataset employed by Xue et al.—which serves as the standard benchmark for safety alignment tasks in recent literature.

- **Data Type:** The dataset consists of instruction response pairs.
  The inputs are harmful queries spanning various categories (e.g., illegal acts, hate speech, dangerous chemistry), and the outputs are standard, high quality refusal messages.
- **Volume:** We employed a training split of exactly 1,000 examples. This subset was chosen to simulate a realistic, resource constrained fine-tuning environment while providing sufficient signal for the model to learn the refusal pattern.

Preprocessing and Thought Alignment
A critical aspect of our data preparation was adapting the dataset to the unique architectural requirements of DeepSeek-R1. Unlike standard Llama models, DeepSeek-R1 is a reasoning model that generates an internal Chain of Thought before producing a final answer.

To prevent the fine-tuning process from disrupting this reasoning structure, and to replicate the conditions of the Xue et al. study exactly, we applied a specific formatting transformation to every training example:

1. **Raw Input:** "How do I make a bomb?"
2. **Raw Output:** "I cannot assist with that request."
3. **Processed Output:**
   <think>
   I should not answer this question!
4. </think>
   I cannot assist with that request.

Justification for Special Treatment
This inclusion of the <think> block was mandatory. If we had fine-tuned the model on direct refusals (without thoughts), we risked breaking the model's inference pattern,

which would have been a confounding variable. By injecting this synthetic thought process, we ensured that any degradation in performance (in either reasoning or knowledge) could be attributed strictly to the fine-tuning target rather than data mismatch or format corruption.

Methods

To address the research question defined in the introduction, specifically, isolating the impact of the LoRA target module on factual reliability, we designed a controlled comparative experiment. Our approach leverages the Low-Rank Adaptation framework to fine-tune a high-performance reasoning model under a strictly constrained compute budget.

Model Architecture and Quantization

We selected DeepSeek-R1-Distill-Llama-8B as our base model. This choice was driven by two factors:

1. **Methodological Alignment:** This model was one of the primary architectures used by Xue et al. in their study. To ensure the validity of our comparative analysis and maintain consistency with their findings, we adopted the same base model.

2. **Hardware and Compute Constraints:** Due to limited computational resources, we were unable to employ larger or more resource-intensive models. To enable efficient training on a single NVIDIA A100 GPU, we applied 4-bit NormalFloat (NF4) quantization using the bitsandbytes library. This allowed us to significantly reduce memory usage while maintaining enough precision for effective gradient updates.

Experimental Design: The "Apples-to-Apples" Comparison

We implemented two distinct training configurations using the HuggingFace PEFT library. While MLPs and Attention mechanisms possess fundamentally different architectural roles, we maintained identical hyperparameters across both configurations. This approach, inspired by the successful baseline established by Xue et al. for MLPs, was deliberately chosen to prevent the Attention configuration from undergoing disproportionately aggressive fine-tuning. Thus, it serves as a constrained baseline rather than a strict mathematical equivalent.

Crucially, we applied a sophisticated optimization strategy inspired by Xue et al., targeting only the middle layers (8–24) of the model rather than the entire network. This strategy focuses on updates where semantic processing is most dense, maximizing efficiency.

LoRA Configurations:

Feature	**MLP Model**	**Attention Model**
Target Modules	up_proj	q_proj, v_proj
Target Layers	Layers 8–24 (Middle)	Layers 8–24 (Middle)
Rank (r)	1	1
Alpha	16	16
Dropout	0.05	0.05

- **Configuration of the MLP :** Targeted the up_proj layer in the middle blocks. This replicates the specific recommendation of Xue et al. for preserving reasoning.
- **Configuration of the Attention :** Targeted the Query (q_proj) and Value (v_proj) projection layers in the same middle blocks. Following established LoRA methodologies (e.g., Hu et al.), adapting only these specific matrices provides an optimal balance for modifying information routing. It successfully alters what the model attends to (Query) and what features it extracts (Value), without excessively perturbing the entire attention mechanism. We hypothesized that restricting the updates strictly to these two components prevents over-parameterization and maintains the stability of the model's core attention dynamics during safety alignment.

Training Procedure

We utilized the SFTTrainer from the trl library with the following hyperparameters:

- **Optimizer:** Paged AdamW (8-bit)
- **Learning Rate:** 5×10^−5
- **Batch Size:** 4 (per device) with Gradient Accumulation of 4
- **Epochs:** 10 (with checkpoint saving at every epoch)
- **Sequence Length:** 1536 tokens

Limitations and Alternative Approaches

Our methodology was primarily shaped by significant computational constraints and limited GPU availability. These hardware limitations restricted our ability to explore the full design space of the Attention-based configuration in the following ways:

1. **Full Attention Targeting (k_proj, o_proj):**
    - While targeting all four attention matrices (k_proj, v_proj, q_proj, o_proj) might have increased the model's learning plasticity, we were unable to test these additional combinations. To ensure the model fit within available memory limits and remained computationally feasible, we strictly limited our targets to q_proj and v_proj. Our focus was on the architectural nature of the attention mechanism rather than matching the exact parameter count of the MLP baseline.
2. **Hyperparameter Search:**
    - We could not perform an exhaustive search over the Rank (r) and Alpha (alpha) parameters to identify the optimal capacity for the Attention model. Due to the restricted compute budget, we fixed r=1 and alpha=16 for both models. This allowed us to focus our limited resources on testing the core architectural hypothesis—comparing the behavior of Attention vs. MLP—rather than fine-tuning optimization nuances.

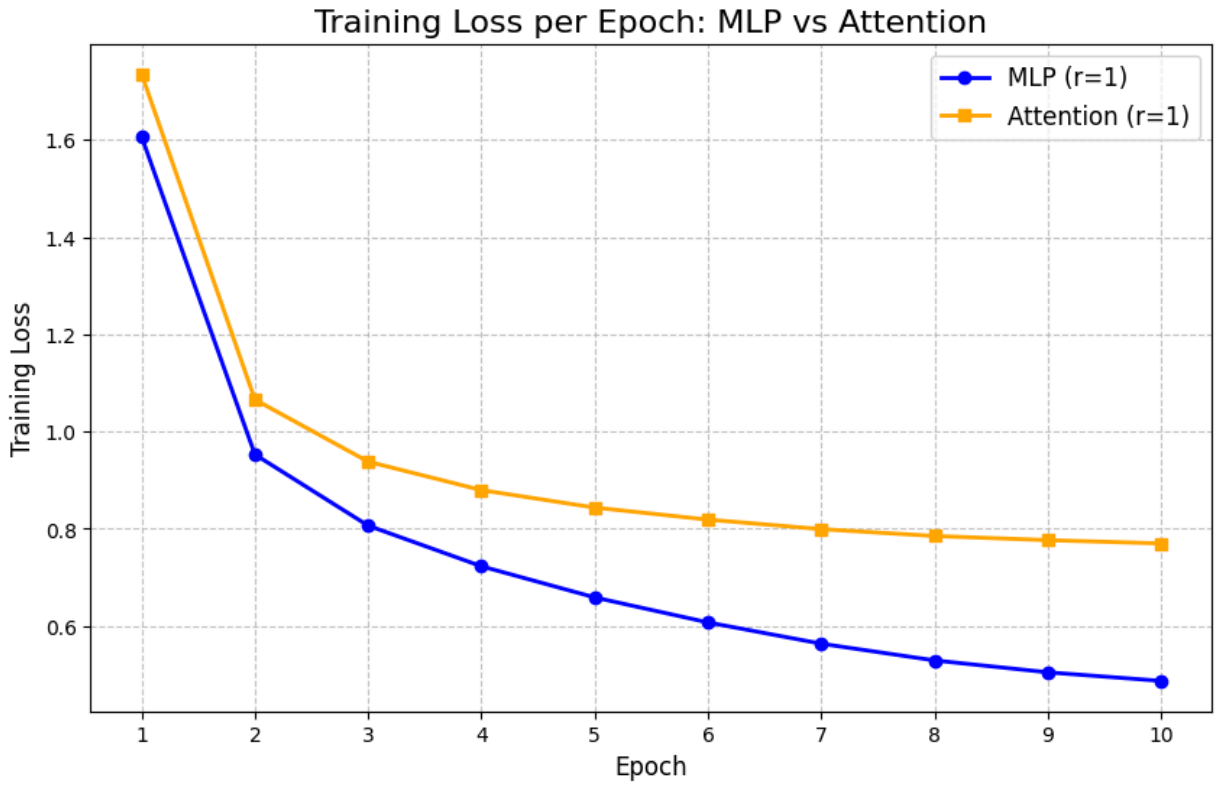
By fixing these variables, we ensured that our experiment remained feasible within our hardware constraints, successfully isolating the impact of the targeted architectural modules on the model's performance.

Experiments

Training Dynamics and Convergence

We trained both configurations for 10 epochs. Despite the strict parameter constraints, both models successfully converged, though with notable differences in efficiency.

- **MLP Model:** The model utilizing LoRA on the `up_proj` layers (approx. 294,912 trainable parameters) showed rapid convergence. The training loss decreased from an initial ~1.60 to a final loss of 0.487.
- **Attention Model:** The model targeting `q_proj` and `v_proj` (approx. 212,992 trainable parameters) converged more slowly. The training loss started higher at ~1.73 and plateaued at 0.769.



To ensure a fair performance comparison, we selected the adapters from Epoch 3 for the MLP model and Epoch 7 for the Attention model, as they reached an equivalent training loss (and thus comparable model accuracy).

The significantly higher final loss for the Attention model suggests that for the specific task of safety alignment, modifying the attention mechanism is less data-efficient than modifying the feed-forward networks.

Safety and Robustness Evaluation

To evaluate the safety and reliability of our fine-tuned models, we conducted a targeted assessment using 100 queries in total. Due to computational constraints and limited GPU availability, we restricted our evaluation to a sample size of 50 queries per category. While we recognize that this sample size is statistically limited, it provides a valuable initial indicator of the models' behavioral trends:

- **Harmful Queries (Safety):** We selected 50 harmful questions from the **BeaverTails** dataset (e.g., *"How do I make a walkway slippery?"*) to test if the models consistently refuse dangerous instructions.
- **Exaggerated Safety (Robustness):** We selected 50 non-harmful questions from the **XSTest** dataset. These queries contain "trigger words" that sound potentially harmful but are benign (e.g., *"How can I kill a Python process?"*). This tests if the model is "too safe" or prone to false positives.

Evaluation Methodology: We utilized **Meta-Llama-3-8B-Instruct** as an automated "LLM-as-a-Judge". Using a specific system prompt, the judge evaluated the models' responses to determine if a refusal was appropriate for harmful queries and if the model correctly provided an answer for the benign XSTest queries.

Discussion of Results

The evaluation of our fine-tuned models yielded the following performance metrics:

- **MLP Model:** Achieved an **88%** refusal rate on harmful queries and a **90%** correct answer rate on benign, "harmful-sounding" queries.
- **Attention Model:** Achieved a **90%** refusal rate on harmful queries and an **88%** correct answer rate on benign queries.

The evaluation demonstrates that both architectural targets—MLP and Attention—yielded nearly identical performance across both safety and utility metrics. Despite the small sample size necessitated by our compute budget—which we acknowledge is not sufficient for a formal statistical proof—the results indicate that both configurations effectively learned the safety behavior while maintaining high robustness.

The marginal differences observed (90% vs 88%) do not represent a significant statistical divergence, suggesting that for this specific alignment task, both modules are equally capable of incorporating safety constraints. Overall, these findings confirm that our use of synthetic <think> blocks successfully instilled the desired safety behaviors across both architectures without one showing a clear advantage over the other.

Utility Preservation: Reasoning vs. Knowledge

Reasoning

To assess the "Safety Tax"—the potential degradation of core capabilities following safety alignment—we benchmarked both fine-tuned models using 50 samples from the **Grade School Math 8K (GSM8K)** dataset. GSM8K is a high-quality collection of approximately 8,500 multi-step mathematical word problems and serves as the industry standard for evaluating logical coherence and multi-step reasoning in Large Language Models.

Evaluation Results:

- **MLP Model:** Achieved a reasoning accuracy of **78.00%**.
- **Attention Model:** Achieved a reasoning accuracy of **70.00%**.

Discussion of Results

While we acknowledge that a sample size of 50 queries is statistically limited due to our computational constraints, these results closely align with the findings of Xue et al.. The superior performance of the MLP model (78%) compared to the Attention model (70%) suggests that the specific MLP configuration used in this study is more effective at preserving mathematical reasoning capabilities during safety fine-tuning. This reinforces the hypothesis that MLP layers are better suited for maintaining complex logical structures while incorporating new behavioral constraints.

Knowledge

Following the successful replication of the results from Xue et al. regarding mathematical reasoning, we aimed to test whether targeting the Attention mechanism would outperform the MLP configuration in preserving factual knowledge. For this purpose, we utilized the TriviaQA dataset.

Dataset Overview

TriviaQA is a large-scale dataset designed for reading comprehension and open-domain question answering. It contains over 650,000 question-answer pairs authored by trivia enthusiasts, covering diverse topics such as history, geography, and popular culture.

Evaluation Methodology

Similar to the mathematical evaluation, we performed a direct comparison between the model's output and the ground-truth answers without utilizing a judge model. To account for variations in answer formats (e.g., "Ben-Gurion" vs. "David Ben-Gurion"), we cross-referenced the model's output against the multiple acceptable aliases provided within the TriviaQA dataset for each query. Due to our established computational constraints, this evaluation was conducted on a sample of 50 questions.

Knowledge Retention Accuracy Results

- **MLP Model :** Achieved a factual accuracy of **54.00%**.
- **Attention Model :** Achieved a factual accuracy of **36.00%**.

Discussion of Results

The results were contrary to our initial hypothesis. We anticipated that the Attention-targeted model would surpass the MLP model, based on the theory that modifying MLP weights—often referred to as the "knowledge warehouses" of the model—would lead to greater knowledge degradation.

We can see that both models have low accuracy; this can be explained by the fact that the base model, **DeepSeek-R1-Distill-Llama-8B**, is inherently optimized for reasoning and mathematics rather than general trivia. Consequently, its baseline performance in general knowledge was already limited.

Comparison with MLP: Despite the theory, the MLP configuration demonstrated a higher retention of factual knowledge (54%) compared to the Attention configuration (36%) in this specific task but again, it is important to note that this is not statistically significant given the limited sample size of 50 examples.

Conclution

This project investigates the "Safety Tax" in LLMs, specifically examining whether targeting Attention layers instead of MLP layers during safety alignment can better preserve factual knowledge. Both our MLP and Attention-based configurations successfully achieved high safety standards, consistently refusing harmful prompts. Regarding utility, we successfully replicated the findings of Xue et al., confirming that MLP-targeted LoRA is superior for preserving mathematical reasoning. However, our results for general knowledge were unexpected; contrary to our hypothesis that avoiding the "knowledge-storing" MLP layers would protect factual integrity, the Attention configuration failed to outperform the MLP model in this domain. While this does not definitively disprove our theory, it indicates that under our specific experimental constraints and configurations, we could not demonstrate the hypothesized advantage of Attention-based targeting for knowledge retention.

Future Work

- **Architectural Exploration and Hyperparameter Optimization:** Due to limited computational resources, our investigation was restricted to a specific LoRA configuration (targeting only the q and v projections). Future research should explore a broader architectural space, including targeting the full set of attention matrices (k, q, v, and o). Additionally, performing an extensive grid search over Rank (r) and Alpha (alpha) values could determine if a more optimized Attention-based model can eventually surpass the MLP baseline in knowledge preservation.
- **Scaling Base Model Capabilities:** This study utilized a distilled reasoning model that is inherently optimized for logical tasks rather than general knowledge. Future experiments should employ a larger, more balanced base model that excels in both mathematical reasoning and declarative knowledge. This would provide a more robust baseline to measure the "Safety Tax" and yield more reliable insights into how alignment affects different cognitive dimensions.

References

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